

SEMESTER-II
ENGINEERING MATHEMATICS-II
SUBJECT CODE: NBS-202

COURSE OBJECTIVE:

The course covers various topics of mathematics required for engineering. The objective of the course is to enhance students' skills and knowledge in mathematical concepts in a way that they can apply effectively in various engineering disciplines.

SYLLABUS:

Unit-I (Partial Differential Equations-I) (8 lectures)

Order and Degree, Formation of PDE, Solution of first order Lagrange's Linear Partial differential equations, Solutions of Non-Linear Partial Differential equations of first order, Charpit's method, Second order PDEs with constant coefficients.

Unit II (Partial Differential Equations-II) (8 lectures)

Classification of Second order partial differential equations, Elliptic, Parabolic and hyperbolic, Separation of variables method to simple problems in Cartesian coordinates including two dimensional Laplace, one dimensional Heat and one dimensional Wave equations.

Unit III (Laplace Transform) (8 lectures)

Laplace transforms and its properties, Unit step Function, Error Function, Inverse Laplace transform, Convolution theorem, Applications for solution of linear and simultaneous differential equations.

Unit IV (Fourier transform and Fourier Series) (8 lectures)

Periodic functions, infinite Fourier transform, properties of Fourier transform, Finite Fourier transform and Fourier Integral theorem, Fourier series, Dirichlet's Conditions, Half range sine and cosine series, Euler's formulae.

Unit-V (Series Solution and Special Functions) (8 lectures)

Series solutions of second order linear differential equations of variable coefficients (Frobenius method), Bessel's and Legendre equations and their solutions, Properties of Bessel functions and Legendre Polynomials.